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Why musical hierarchies?

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Abstract

Credible signaling may have provided a selection pressure for producing and discriminating increasingly elaborate proto-musical signals. But, why evolve them to have hierarchical structure? We argue that the hierarchy of tonality and meter is a byproduct of domain-general mechanisms evolved for reasons other than credible signaling.

The target article by Mehr and colleagues provides a welcome critique of prevailing evolutionary theories of music while also advancing their own credible signaling proposal. We find many aspects of this promising. However, although adaptations for rhythm and melody seem plausible, we take issue with the claim that credible signaling resulted in a “grammar-like, combinatorially generative interface” based on the “hierarchical organization of meter and tonality.”

Coalition signaling provides plausible reasons to evolve the capacity to produce and discriminate rhythmically coordinated displays. This is supported in the cited data on birds (Hall & Magrath, 2007; Tobias et al., 2016) and primates (Geissmann, 2000). But, it does not, as far as we can see, provide reasons to evolve hierarchical means of doing so. And indeed, these data only show evidence of

rhythmic coordination in terms of temporal precision or synchronization and provide no evidence for or against hierarchy.

Similarly, parent–infant signaling provides evolutionary reasons for melodic signals and infant sensitivity to them. A comparison of animal contact calls (Bouchet, Blois-Heulin, & Lemasson, 2013; Leighton, 2017) and data on genomic imprinting disorders in humans (Mehr, Kotler, Howard, Haig, & Krasnow, 2017) supports these claims. But, here too: why hierarchies? For the purpose of signaling attention to an infant, or for contact calls more generally, hierarchical organization poses no obvious advantage. There is also limited evidence for contact calls being hierarchically organized. Moreover, although some brain areas show differential responses to tonal structure from birth (Perani et al., 2010), behavioral sensitivity only begins to manifest at around 4 years of age before continuing to develop into the teenage years (Brandt, Gebrian, & Slevc, 2012; Corrigan & Trainor, 2014).

Taken together, although the hierarchical properties of meter and tonality are a design feature of the musical capacity, their presence is not so clearly motivated by credible signaling.

Hierarchies, however, are not unique to music. They are found in other cognitive domains such as language (Chomsky, 1957), vision (Bill, Pailian, Gershman, & Drugowitsch, 2020), metacognition (Frith, 2012), and action planning (Miller, Galanter, & Pibram, 1960). In nonhuman primates, they are found in social learning (Byrne & Russon, 1998) and tool use (Byrne, Sanz, & Morgan, 2013; Greenfield, 1991). Musical hierarchy may, therefore, be better conceived as using generic mechanisms evolved for reasons other than as a specific adaptation for credible signaling.

We have previously argued that the hierarchy of both musical and linguistic structures derives from mechanisms originally evolved for action planning (Asano & Boeckx, 2015; see also: Fitch & Martins, 2014; Jackendoff, 2009). The inspiration for much of this thinking was Karl Lashley's (1951) prescient insight that complex actions generally, and those for music and language specifically, control their sequential manifestation through hierarchical plans. Doing so, he argued, was necessary for flexibility and robustness, especially for more complex and abstractly motivated actions in which the limitations of control by linear associative chaining are laid bare.

The primary neurocognitive mechanism underlying this capacity is hierarchical cognitive control and comprises a combination of executive functions (maintenance, selection, and inhibition). Maintenance is subserved by prefrontal areas (together with their parietal connections) and selection and inhibition by the basal ganglia. The orchestration of these functional areas through a number of distinct cortico-basal ganglia-thalamocortical circuits enables complex and flexible behavior (Badre & Nee, 2018). Consistent with Lashley's insight, these neural circuits are not only implicated in action planning, but also for processing musical and linguistic hierarchies (Asano et al., 2021; Fitch & Martins, 2014; Jeon, Anwender, & Friederici, 2014; Slevc & Okada, 2015).

Functional explanations of behavior are essential for understanding biological evolution. But, based on these alone, the determination of how they are translated into mechanisms is too underconstrained. Are new mechanisms evolved *de novo*? Or are existing ones tweaked and put to new use? And then how may these and other proximate mechanisms in turn constrain the space of ultimate reasons that guides selection in a reciprocal cycle? (Laland, Sterelny, Odling-Smee, Hoppitt, & Uller, 2011). As Tinbergen (1963) suggested, the biological study of behavior (and cognitive systems, in the current paper) should give equal attention to each of four questions: mechanism, ontogeny, phylogeny,

and function. Each provides unique constraints whose combined consilience is the basis for robust theory.

One notable “so what?” of all this for the target article is that adaptations for credible signaling may also have implications for language. According to our proposal, the structural complexity of both music and language partly derives from generic hierarchical cognitive control mechanisms that interface with auditory and motor systems. Compared to nonhuman primates, humans have substantially greater white-matter connectivity both within the hierarchical control circuits and through the dorsal auditory pathway that links motor, auditory, and parietal areas with the prefrontal cortex (Barrett et al., 2020; Rilling et al., 2008). Adaptations for producing and perceiving rhythmically coordinated audio-motor displays and for fine-scale vocal control of pitch conceivably include an expansion of this shared connectome (Merchant & Honing, 2014; Patel & Iversen, 2014), thus entangling the evolution of both domains. This would also be consistent with claims about music-to-language transfer effects more generally in ontogeny (Patel, 2011; Zatorre, 2013).

To conclude, the credible signaling proposal of Mehr and colleagues is commendable. But, we suggest that it can be further improved by considering interactions of proximate and ultimate causes, and specifically how this may clarify the origins of musical hierarchies.

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A neurodevelopmental disorders perspective into music, social attention, and social bonding

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Abstract

Our commentary addresses how two neurodevelopmental disorders, Williams syndrome and autism spectrum disorder, provide novel insights into the credible signaling and music and social bonding hypotheses presented in the two target articles. We suggest that these neurodevelopmental disorders, characterized by atypical social communication, allow us to test hypotheses about music, social bonding, and their underlying neurobiology.

Neurodevelopmental disorders characterized by atypical social profiles provide a unique window into the relationship between sociability, music, and the evolutionary origins of music. Here, we focus on two neurodevelopmental disorders characterized by atypical development of social communication – Williams syndrome and autism spectrum disorder (ASD). We (1) provide additional consideration about music as a credible signal for parent–infant interactions (Mehr, Krasnow, Bryant, & Hagen, target article) and (2) extend neurobiological predictions about music and social bonding (MSB; Savage et al., target article).

Individuals with Williams syndrome, a genetic neurodevelopmental disorder caused by the deletion of ~28 genes on chromosome 7, exhibit marked hypersociability accompanied by difficulties in social pragmatics (Barak & Feng, 2016). Individuals with ASD, an etiological heterogeneous set of neurodevelopmental disorders, exhibit deficits in social communication and interactions (Barak & Feng, 2016). Music is a relative strength and interest in both Williams syndrome and ASD. Individuals with Williams syndrome exhibit increased emotional responsiveness to music and increased neural responses to musical stimuli compared to typical peers (Kasdan, Gordon, & Lense, 2020; Lense, Gordon, Key, & Dykens, 2014), although behavioral studies of musical skills show mixed findings and substantial individual differences (Hopyan, Dennis, Weksberg, & Cytrynbaum, 2001; Martens, Reutens, & Wilson, 2010). Individuals with ASD generally exhibit age-appropriate melody and rhythm skills (Jamey et al., 2019; Tryfon et al., 2017) and better pitch interval detection compared to controls (Heaton, 2005). Additionally, individuals with ASD show preserved emotional processing of music at both behavioral and neural levels, despite overall impairments in emotion recognition (Caria, Venuti, & De Falco, 2011; Molnar-Szakacs & Heaton, 2012).

As in typical development, music is used in these populations to leverage social attention and bonding, including during parent–child interactions (Steinberg, Shivers, Liu, Cirelli, & Lense, 2020). Because of impaired social communication of children with ASD and Williams syndrome, parents adapt and increase their social signaling in order to capture and modulate their children's attention and emotions; this frequently involves adapting the “musicality” of the social signal such as through increased rhythmic predictability and pitch contours (e.g., increased use of infant-directed speech; Cohen et al., 2013; Quigley, McNally, & Lawson, 2016). Music and song activities provide an ecologically valid means for scaffolding parental signaling during these shared social interactions (Lense & Camarata, 2020). This is consistent with and extends the idea of music as a credible signal for parent–infant interactions (Mehr et al.). The core features of music (e.g., discrete pitches and predictable and hierarchical rhythms) incorporated into both credible signaling and MSB hypotheses can predictably, reliably, and efficiently structure social rhythmic signals such as vocalizations, movement, gaze, and touch (Lense & Camarata, 2020).